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Multimedia Portfolio Behavior Analysis Based on Bayesian Game Algorithm

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Abstract
A portfolio prediction model based on Bayesian game algorithm is proposed to improve the accuracy of correlation analysis model in portfolio application. Firstly, the portfolio problem is analyzed. The market value constraint and the upper bound constraint are combined considering the general portfolio model according to Markowitz theory, it improves the portfolio model and obtains the portfolio model with mixed constraints. Secondly, a probabilistic portfolio monitoring strategy based on incomplete information non-cooperative Bayesian game is proposed. The framework combines the enterprise investment behavior detection module with the specified rules to identify the portfolio pattern, the interaction between portfolio selection and monitored investment behavior is modeled as a two-person non-cooperative Bayesian game, which allows portfolio selection to adopt a probability monitoring strategy based on game Bayesian Nash equilibrium, thereby reducing the computational complexity of the portfolio approach introduced into the game algorithm. Finally, the effectiveness of the proposed algorithm is verified by simulation experiments.

Keywords Bayesian · Game algorithms · Enterprise investment · Non-cooperation · Investment portfolio

1 Introduction

The core issue of quantitative investment is to accurately forecast the changes in assets prices [2]. If the rise and fall of asset prices can be forecasted, then investment managers can build a more effective portfolio to obtain greater investment returns [13]. Under the traditional mode of "analyst-fund manager", analysts forecast the future assets prices through personal subjective analysis, but the mode is faced with two challenges: First, the sharp increase in the number of assets makes the traditional model require a large implementation costs, which makes the model no longer applicable; secondly, the analyst's subjective judgment is influenced by his/

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her emotions, which makes his/her forecasts no longer accurate [1, 4, 10]. The quantitative investment model can solve these two challenges [5, 8, 9]. A quantitative investment model is established in this paper as an attempt to solve issue of forecasting assets prices and investing in securities.

Game learning (1) By learning the reproducible mode from a large number of assets and using the learned mode for forecasts? Because the game learning algorithm has high forecast accuracy and good generalization ability, the excellent forecast ability of the model can be applied to the off-sample assets. This technology is widely used in quantitative investment, such as risk forecast, etc. Technical analysis is widely used by market practitioners for forecasting future asset prices in actual transactions [3]. Compared with fundamental assets, technical indicators can be easily obtained from market assets and have strong timeliness [6]. Existing technical analysis studies usually simulate a single technology indicator or technical graphic to verify whether a technology is effective, but the studies on multiple technical indicators or technical analysis methods are relatively in shortage [11]. Theoretically, effectively mining the information of multiple technical analysis methods can bring about better forecast result than mining a single technical analysis method [7]. In order to better mine the information in technical analysis, a set of quantitative investment algorithm is designed in this paper based on game learning and technical indicators [12]. Several common technical indicators are mined in this model by game learning algorithm so as to forecast the ups and downs of the stock prices after a few days, and build the portfolio according to the forecast direction. Based on the difference in the game study technology adopted, the game algorithm in this paper covers three strategies, which respectively are the strategy based on neural network and technology analysis, the strategy based on support vector machine and technology analysis, and the strategy based on neural network and technical analysis integrated by Adaboost (the algorithm in this paper).

2 Portfolio analysis model

2.1 Effective boundary of combination model

Investors need to consider the following steps in their portfolio decision-making [11~12]: (1) use the selected stock securities to solve the effective portfolio and obtain the corresponding portfolio weight; (2) make analysis on the return rate expectation and portfolio variance to obtain the weight calculation value of the portfolio; (3) make analysis on the points on the boundary of portfolio analysis model to obtain the weight calculation value of the portfolio, wherein, the most critical is to calculate the effective value of the portfolio boundary in the following model (1), and obtain the management weight optimization of the portfolio:

$$\begin{cases} \min f_1 = x' \sum x \\ \max f_2 = \mu' x \\ s.t. \quad x \in S \end{cases} \tag{1}$$

In the formula, $\sum$ is the covariance matrix of the portfolio problem, x is the weight vector of each model of the portfolio problem, S is the research feasible domain of the portfolio problem, and μ is the expectation of the yield. The traditional calculation methods for portfolios include: constraint method, weight summation method, quadratic programming optimization

calculation method and so on. Quadratic programming optimization calculation method is adopted in this paper as the calculation method of exact boundary.

2.2 Quadratic programming optimization method

Simple multiple quadratic programming is usually adopted in the above algorithm g to achieve management weight optimization of the portfolio. What is obtained by such methods is not the exact boundary of the portfolio problem, but only the rough calculation result of the effective boundary [12]. Markowitz designed a key line solving method through the research and analysis, analyzed the portfolio model (1) by weighting method, retained the weight parameters in the model, and changed the calculation method of taking the parameters as pre-variables in the classical algorithm. This processing method is called quadratic programming method.

Compared with traditional algorithms such as constraint method and weight summation method, the quadratic programming algorithm can achieve an accurate solution of the effective boundary instead of a rough solution. At the same time, the biggest advantage of the quadratic programming algorithm is that it can reveal the effective combination of the hyperbolic front and back connection of the portfolio and accelerate the calculation process.

2.3 Model design

In order to achieve the performance analysis of calculation process by the particle swarm algorithm in the portfolio model, the two key indicators of portfolio constraints and the number of portfolios are respectively studied in this paper. The number of stocks grouped is $n = [5, 10, 50, 100, 200, 300, 400]$. Then the different constraints are used to set, the particle swarm optimization algorithm and quadratic programming algorithm are adopted in this paper to solve the effective boundary of the portfolio problem respectively, the effective evaluation index of the particle swarm optimization algorithm is obtained by this comparison method. The specific flow is shown in the algorithm block diagram in Fig. 1.

Group I: the selection of a standard model for portfolio issues is as follows:

$$\begin{cases} \min f_1 = x' \sum x \\ \max f_2 = \mu' x \\ s.t. \quad 1' x = 1; x \geq 0 \end{cases} \tag{2}$$

Group II: the upper bound constraint is applied to the standard portfolio selection problem in formula (2), and the combined improvement model can be obtained, as shown in formula (4); wherein, u is the upper bound of the standard portfolio model, 1 is the vector with element being 1, and 0 is the vector with element being 0:

$$\begin{cases} \min f_1 = x' \sum x \\ \max f_2 = \mu' x \\ s.t. \quad 1' x = 1; 0 \leq x \geq u \end{cases} \tag{3}$$

In this study, stocks in the quantity of 5 and 10 are studied, and the upper bound in the two groups is set to be 0.4, and the upper bounds in the grouped stocks in the other groups are set to be 0.1.

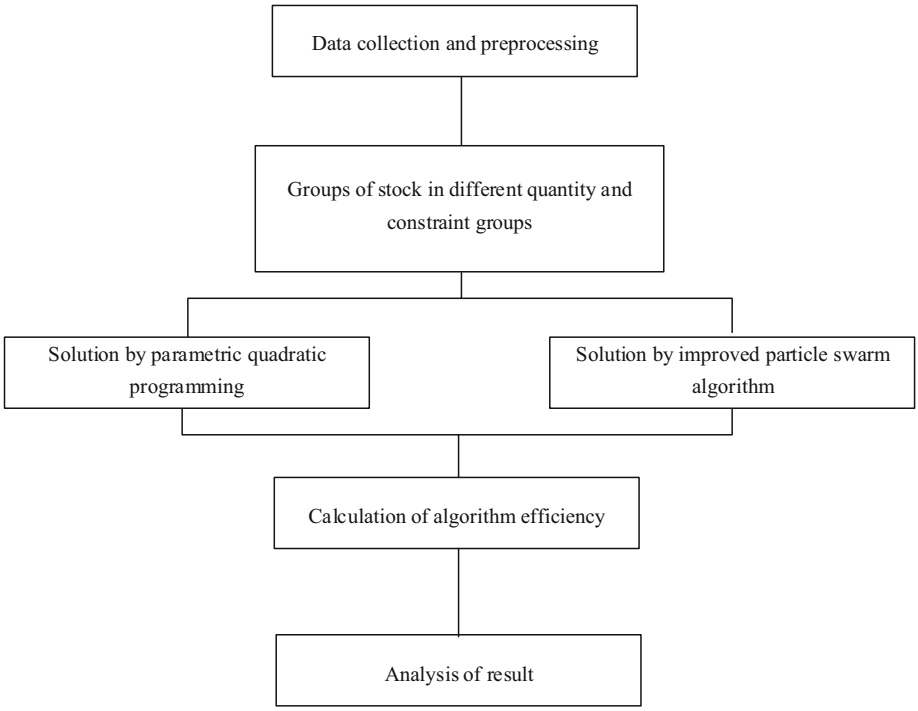

Fig. 1 Algorithm Efficiency Evaluation Process

Group III: as shown in formula (5),the market value constraint is applied so as to improve the standard portfolio model shown in formula (1), wherein, the parameter m is the stock market value in the selected research object, and it is a numerical vector.

$$\begin{cases} \min f_1 = x'\sum x \\ \max f_2 = \mu' x \\ s.t. \quad 1'x = 1; m'x \geq m \end{cases} \tag{4}$$

In the research of this paper, for the study of the market value vector of the target, the average market value of the stocks it represents on the stock market is selected.

3 Bayesian game model for detection of enterprise portfolio

3.1 Definition of game matrix

The amount of information exchanged with the financial market is required to be minimized to preserve the energy for portfolio approach agents. In addition, the game algorithm is characterized by unstable investment subjects. Therefore, the excessive calculation complexity associated with enterprise portfolio detection may result in low calculation accuracy in the game algorithm. In order to solve these problems, we proposed probabilistic monitoring strategy for portfolio approach agents based on game theory Nash Equilibrium (NE). This allows portfolio approach agents to minimize the calculation complexity capacity introduced

into the game algorithm. The interaction between portfolio approach agent and investment behavior is expressed as a dynamic Bayesian game of two players, wherein, the player P_i is a potential investment subject (malicious investment subject) and the other player P_j is a defender (portfolio approach agent). The type $P_i(\theta_i)$ is private information that is unknown by P_j. If $\theta_i = 1$ (or 0), P_i is normal (or malicious).

The defensive role P_j is always normal ($\theta_j = 1$), and has two pure strategies: {monitoring strategy, non-monitoring strategy}. Because the defensive role is not sure of the type of opponent's role (malicious or normal), the interaction between the role P_i and the role P_j can be expressed as an incomplete information Bayesian game $G = \langle N, S, U \rangle$, wherein: $N = \{P_i, P_j\}$ is the decision-making opponent of the game. $S = S_i \times S_j$ is the game strategy space of S_i and S_j.$U = U_i \times U_j$ is the utility payment corresponding to the policy space. U_i and U_j respectively are the rewards of the player P_i and the player P_i corresponding to the strategic space S_i and S_j.

Suppose that the detection rate (DR) and false positive rate (FP) of the defensive role P_j are expressed by α and γ, wherein, $\alpha, \gamma \in [0, 1]$. C_{a_k} and C_{m_k} are respectively used for expressing the costs involved in initiating investment and monitoring investment behavior n_k. Implicitly assume that C_{a_k}, $C_{m_k} \ll w_k$, otherwise the investing subject and the defender will have no effective incentive to invest in and defend against the investment subject n_k. Tables 1 and 2 respectively show the game matrices corresponding to the interactions between the portfolio approach agent and investment behavior when the type of investment subject n_k is malicious and normal. The goal behind the design of these game matrices is to enable portfolio approach agents to maximize the probability of their successful detection of portfolio mode while allowing the portfolio approach to minimize this maximization goal.

3.2 Bayesian game process

This non-cooperative game between portfolio approach agents and investment behavior can be modeled as an incomplete information Bayesian game. Their payment is adjusted according to natural movements by adding a virtual player called Nature (expressed by N). Calculation is made as follows in the process of Bayesian game.

Process 1: If the type of investment subject n_k is malicious and if it plays its pure strategy of "investment", then the cost of the portfolio approach agent when playing its pure strategy "monitor" and "non-monitor" is as follows:

$$U_{Agt}(Monitor) = p_o((2\alpha - 1)w_k - C_{m_k}) - (1 - p_0)(\gamma w_k + C_{m_k}) \tag{5}$$

$$U_{Agt}(Not\ Monitor) = -p_o w_k \tag{6}$$

Table 1 Game Matrix When the Investment Subject n_k Is Malicious

Type	Monitoring	Non-monitoring
Investment	$(1-2\alpha)w_k - C_{a_k}$ $(2\alpha-1)w_k - C_{m_k}$	$w_k - C_{a_k}, -w_k$
Non-investment	$0, -\gamma w_k - C_{m_k}$	0, 0

Table 2 Game Matrix of Normal Investors

Type	Monitoring	Non-monitoring
Non-investment	0, $-\gamma w_k - C_{m_k}$	0, 0

The payment utility function "$U_{Agt}(Monitor)$"indicates that the gained obtained by the portfolio approach agent through monitoring operation and preventing costs associated with asset theft, as well as through successfully detection of damage to the asset integrity with the assets implemented by the malicious investment subject n_k. Similarly, the payment utility function "$U_{Agt}(Not\ Monitor)$"indicates the loss suffered by the portfolio approach agent in the event of asset theft, and the loss of asset integrity when the malicious investment entity initiates the investment and the portfolio approach agent does not perform any monitoring operations.

Process 1: When a portfolio mode agent executes its pure strategy "monitor", the expected return of a malicious investment subject that performs n_k its pure strategy of "investment" and "non-investment" is:

$$U_{n_k}(Attack) = p_o((1-2\alpha)w_k - C_{a_k}) \tag{7}$$

$$U_{n_k}(Not\ Attack) = 0 \tag{8}$$

The utility function "$U_{n_k}(Attack)$" indicates the gain that the malicious investment subject obtains when it initiates an investment based on successful assets destruction and investor subject compromise. Similarly, the utility function "$U_{n_k}(Not\ Attack)$"indicates the loss suffered when no investment is performed.

Process 3: Therefore, if $U_{Agt}(Not\ Monitor)U_{Agt}(Monitor)$>, which means, if $p_0 > \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$, when the malicious investment subject n_k plays its role of pure strategic investment, then portfolio approach agent will play its role of pure strategy "monitor". However, when the portfolio approach agent uses its pure strategy "monitor", the natural response of a malicious investment subject n_k will be the adoption of its pure strategy of "not investing". Therefore, when $p_0 > \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$, the strategy (malicious investment, no investment under normal circumstances) is not BNE. However, if $U_{Agt}(Monitor) < U_{Agt}(Not\ Monitor)$, i.e. $p_0 < \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$, regardless of the strategy of the malicious investor n_k, the portfolio approach agent will play its role of pure strategy "non-monitor". Therefore, when $p_0 < \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$, ((invest) = malicious, "not invest" = normal), "not monitor", "p_0") is a pure BNE strategy.

Process 4: If the player uses his pure strategy of "not investing", then the main strategy of the player P_j will be the adoption of the strategy of "not monitoring" regardless of the value of p_o. However, if the player P_j adopts their pure strategy of "not monitoring", the natural reaction of the malicious player P_i will be the adoption of his pure strategy of "investing". Therefore, the strategy ("invest" = malicious, "no invest" = normal), "no monitor" is not a BNE strategy.

Through the analysis of different combinations of strategies, it is found that there is no pure strategy BNE when $p_0 > \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$. However, any non-cooperative game consists of a set of roles and strategies that always have a mixed strategy Nash equilibrium. Therefore, in this case, the game of the hybrid strategy BNE can be obtained. If the type of investment subject is

malicious, let the investment subject n_k use its probability p to execute its pure strategy of "investing". In this case, the expected return for the portfolio approach agent to play its role of pure strategy "monitor" is as follows:

$$\begin{aligned} U_{Agt}(Monitor) &= pp_o((2\alpha-1)w_k-C_{m_k}) \\ &-(1-p)p_o(\gamma w_k + C_{m_k})-(1-p_o)(\gamma k_k + C_{m_k}) \end{aligned} \tag{9}$$

The role of portfolio approach agents in its pure strategy "non-monitoring" is:

$$U_{Agt}(Not\ Monitor) = -pp_0 w_k \tag{10}$$

On the other hand, when the portfolio approach agent plays its strategy "monitor" at probability q, the proceeds of the malicious investment entity n_k playing its pure strategy of "investing" and "not investing" are as follows:

$$\begin{aligned} &U_{n_k}(Attack) = \\ &p_o(q((1-2\alpha)w_k-C_{a_k}) + (1-q)(w_k-C_{a_k})) \end{aligned} \tag{11}$$

$$U_{n_k}(Not\ Monitor) = 0 \tag{12}$$

According to the equation $U_{Agt}(Monitor) = U_{Agt}(Not\ Monitor)$, $p = \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k p_o}$ can be obtained, and it is the probability that the malicious investor subject n_k will "invest" according to its strategy under the BNE of the game. Similarly, through $U_{n_k}(Attack) = U_{n_k}(Not\ Attack)$, it can be determined that the probability that the agent will perform the monitoring strategy under the BNE is $q = \frac{w_k - C_a}{2\alpha w_k}$. Therefore, when $p_o > \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k}$, there is no pure strategy NE, there is a mixed strategy BNE corresponding to the strategy combination, wherein, $p = \frac{\gamma w_k + C_{m_k}}{(2\alpha+\gamma)w_k p_o}$, and $q = \frac{w_k - C_{a_k}}{2\alpha w_k}$.

After the analysis of the BNE strategy, it is found that the high malicious confidence value of the portfolio approach agent to the portfolio approach n_k leads to a significant reduction (p) in the investment probability of the investment subject n_k. However, the disadvantage of the Bayesian game framework is the need to determine the opponent's malicious confidence value (p_o) in advance. To solve this problem, the interaction between the portfolio approach agent and the investment behavior is expressed as a multi-stage non-cooperative game, in which the following Bayesian rules are used to update the malicious belief of investment behavior (n_k) in the game stage:

$$\begin{aligned} &p_j(\theta_i|a_i(t_k), a_i(t_{k-1})) \\ &= \frac{p_j(\theta_i|a_i(t_{k-1}))P(a_i(t_k)|\theta_i, a_i(t_{k-1}))}{\sum_{\tilde{\theta}_i} p_j(\tilde{\theta}_i|a_i(t_{k-1}))P(a_i(t_k)|\tilde{\theta}_i, a_i(t_{k-1}))} \end{aligned} \tag{13}$$

In the formula, $P(a_i(t_k)|\,\theta_i, a_i(t_{k-1}))$ is the probability that the investment subject will function in the kth stage. Because its type is θ_i, and its observational effect in the $(k-1)$ stage is $a_i(t_{k-1})$. Set the detection rate and FP rate of the jth portfolio agent to be α_j and γ_j respectively. The conditional probability given by the above Bayesian formula can be calculated by the following equation:

$$P(a_i(t_k)|\theta_i, a_i(t_{k-1}))a_i(t_k)(k-1)a_i(t_{k-1})\gamma_j \tag{14}$$

$$P\Big(a_i(t_k) = Attack|\theta_i = 1, a_i(t_{k-1})\Big) = p\alpha_j + (1-p)\gamma_j \tag{15}$$

$$P(a_i(t_k) = Attack|\theta_i = 0, a_i(t_{k-1})) = \gamma_j \tag{16}$$

$$P(a_i(t_k) = Not\ Attack|\theta_i = 0, a_i(t_{k-1})) = 1-\gamma_j \tag{17}$$

4 Experimental analysis

In order to verify the effectiveness of the game algorithm model in this paper, 846 target stocks available for securities margin trading were selected to see their daily assets from January 2010 to May 2015 with a total of 1432 trading days. In order to avoid the impact brought by ex-dividend and ex-right, and dividend distribution, the price assets are the one-day trading volume of the price assets and the underlying securities after ex-dividend and ex-right. All assets are taken from CSMAR asset repository. The statistic of coefficient of linear decision function is listed in Table 3 according to the contribution situation. It can be found that although the linear SVM integrates 19 technical indicators in a linear combination to predict the stock price fluctuation in the next few

Table 3 Analysis on the Importance of Technical Indicators in Decision Function

Indicator weighting	Technical indicator	Average hammer	Standard deviation	Minimum value	Median value	Maximum value
1	OBV	−0.1613	0.8035	−1.7820	−0.0933	1.8080
2	NVI	0.1221	0.8159	−1.8008	0.0467	1.7591
3	PVI	−0.0867	0.5702	−1.3138	−0.1556	1.1206
4	CMO	−0.0847	0.8961	−2.2215	−0.0673	1.9505
5	K value	−0.0673	1.1340	−1.7851	−0.2939	2.0113
6	MTM	0.0609	0.3004	−0.7657	0.0269	0.6758
7	VR	−0.0565	0.5598	−1.5341	−0.0586	1.3871
8	AMA	0.0536	0.4514	−1.1670	−0.0305	1.1661
9	RVI	0.0521	1.0016	−1.8862	0.0505	1.8899
10	PSY	0.0395	0.4288	−1.1103	0.0269	0.9729
11	j value	0.0394	0.5768	−1.0923	0.1949	1.2115
12	D value	−0.0373	0.6297	−1.0709	−0.1278	1.1001
13	TRIX	0.0278	0.3028	−1.0100	0.0284	0.7524
14	MACD	0.0275	0.5231	−0.9771	0.0883	1.0632
15	DPO	0.0273	0.6135	−1.3516	−0.0353	1.6380
16	VHF	−0.0254	0.2021	−0.4866	−0.0286	0.5020
17	ROC	−0.0222	0.8911	−1.6129	0.2790	1.6965
18	BIAS	−0.0089	0.4878	−1.2486	−0.0506	1.3019
19	Forcelndex	0.0086	0.2857	−0.9489	0.0059	1.0042

days, the characteristics of each technical indicator play different roles in the decision-making. Among the total of 19 technical indicators, ten of them have the synclastic influence, which means, the increase in the index value is in the same direction with the increase in future stock; the other nine are have the reversed influence, which means, the increase of the index value is in the opposite direction with the increase in future stock price, or the same direction as the decline. The most influential 5 indicators are OBV indicator (reversed influence), NVI indicator (synclastic influence), PVI indicator (reversed influence), CMO indicator (reversed influence) and K value (reversed influence). The least influential five indicators are Force-Index (synclastic influence), BIAS (reversed influence), ROC indicator (reversed influence), VHF indicator (reversed influence) and DPO indicator (synclastic influence).

Figure 2 shows the relationship between the integration times and the forecast accuracy of the algorithm model in this paper, which is consistent with the theoretical expectation, and the accuracy forecast by the strategy increases with the integration times. For example, the proceedings through neural network integrated for 10 times are significantly higher than those through the single neural network strategy. At the same time, although integrated learning can improve the predictive performance of neural networks, the increase is limited. After the integration times are more than 15 times, the forecast accuracy enters a plateau, stable at 60%. In short, through integration learning, the forecast power of ordinary neural network strategy can be improved by about 4%, indicating that the effect of integration learning is significant. Length of the training set: When extending the training interval, more information can be used in the model training and the effect of the classifier will be enhanced theoretically. Figures 2-3 are the curve of change in the annualized rate of return by the game algorithm strategy after changing the length of the training set. It can be seen from the figure: on the one hand, when the training set increases in the algorithm in this paper, the accuracy rate rises first, then maintains at a certain platform, and finally declines; on the other hand, the performance of SVM-TEA strategy is in line with expectations, with the training set increases, the prediction accuracy rate rises, but the rate of increase tends to be gentle.

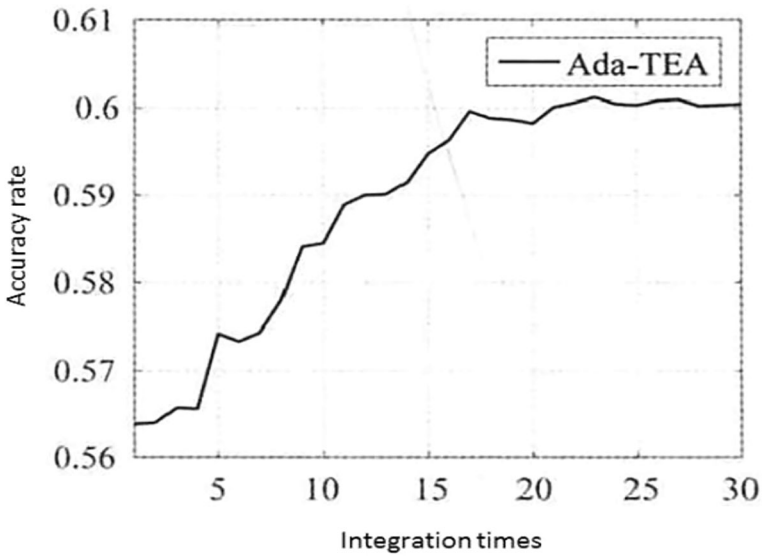

Fig. 2 The influence of Integration Times on the Forecast Accuracy of the Model

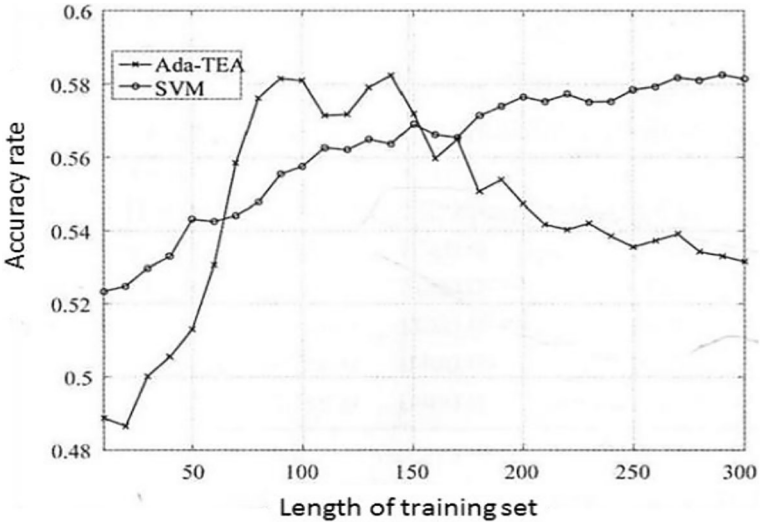

Fig. 3 The Influence of Length of Training Set on the Forecast Accuracy of the algorithm in This Paper and SVM-TEA Model

5 Conclusion

Three kinds of game learning algorithms, including support vector machine and neural network algorithm are adopted in this paper. With technical indicators as the input of the algorithm the forecast of future asset price fluctuations is output, and the portfolio is constructed through the forecasted rise and fall. Empirical results show: First of all, the game algorithm strategy in this paper is superior to other strategies (classical technical analysis strategy, linear regression strategy and buy-and-hold strategy) in terms of risks and benefits, and the forecast accuracy is over 55%, significantly higher than the random probability of 50%. Secondly, through adoption of the algorithm, the forecast accuracy of the neural network strategy can be significantly improved, and at the same time the benefits of the construction strategy are also significantly improved and tend to be stable after the integration times increase. Finally, in the sensitivity analysis of the model, the tolerances of three game studies on transaction costs are much higher than the actual transaction costs of the market, wherein, under the cost of 0.15%, the returns of more than 20% can still be obtained by adopting the strategy based on support vector machine.

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